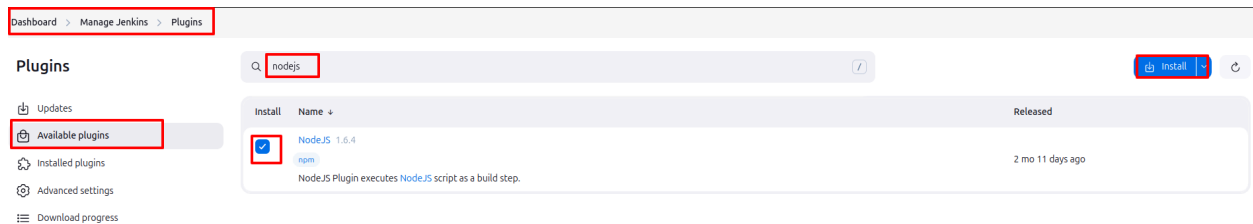


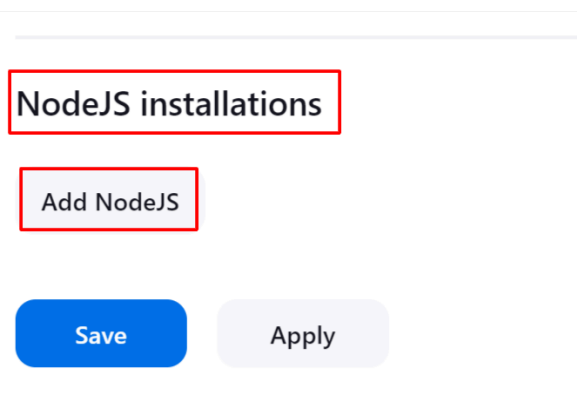


Lab 4S. Creating A Pipeline for the NodeJS Result Application (Solution)

Install the **NodeJS** plugin for Jenkins.



From **Jenkins > Manage Jenkins > Tools > NodeJS Installations**, add a NodeJS installation with version 22.4.0.



Name the installation exactly **NodeJS 22.4.0** as the name will be referred to later in the pipeline.

NodeJS

Name

NodeJS 22.4.0

☒ Install automatically ?

Install from nodejs.org

Version

NodeJS 22.4.0

For the underlying architecture, if available, force the installation of the 32bit package. Otherwise the build will fail

☐ Force 32bit architecture

Global npm packages to install

Specify list of packages to install globally -- see npm install -g. Note that you can fix the packages version by using the syntax 'packageName@version'

Create a new job inside the **instavote** folder called **result-build**, this time as a freestyle project.

Dashboard > instavote > New Item

New Item

Enter an item name

result-build

Select an item type

☒ Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

☐ Maven project
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.

☐ Pipeline
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

Paste in the **example-voting-app** Git repository that you forked in the **Source Code Management** section.

The screenshot shows the Jenkins Configuration page for the 'instavote' job. The 'Source Code Management' tab is selected in the left sidebar. The 'Git' radio button is chosen under 'Source Code Management'. The 'Repository URL' field contains 'https://github.com/okapetarios/example-voting-app'. The 'Credentials' dropdown is set to '- none -'. The 'Branches to build' section has the 'Branch Specifier (blank for 'any')' set to '*/master'. The 'Repository browser' is set to 'Git'. The 'Save' button is highlighted in blue.

Dashboard > instavote > result-build > Configuration

Configure

- General
- Source Code Management**
- Triggers
- Environment
- Build Steps
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ **Git**

Repositories

Repository URL: `https://github.com/okapetarios/example-voting-app`

Credentials: `- none -`

+ Add

Advanced

Add Repository

Branches to build

Branch Specifier (blank for 'any'): `*/master`

Add Branch

Repository browser: `Git`

Save Apply

Select **Poll SCM** with an interval of 2 mins by pasting `H/2 * * * *` into the **Schedule** input box.

Dashboard > instavote > result-build > Configuration

Configure

- General
- Source Code Management
- Triggers**
- Environment
- Build Steps
- Post-build Actions

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- ☐ Trigger builds remotely (e.g., from scripts) ?
- ☐ Build after other projects are built ?
- ☐ Build periodically ?
- ☐ GitHub hook trigger for GITSCM polling ?
- ☒ Poll SCM ?

Schedule ?

`H/2 * * * *`

Would last have run at Saturday, May 3, 2025, 1:39:31 AM Coordinated Universal Time; would next run at Saturday, May 3, 2025, 1:39:31 AM Coordinated Universal Time.

☐ Ignore post-commit hooks ?

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

- ☐ Delete workspace before build starts
- ☐ Use secret text(s) or file(s) ?
- ☐ Provide Configuration files ?
- ☐ Add timestamps to the Console Output
- ☐ Inspect build log for published build scans
- ☐ Provide Node & npm bin/ folder to PATH
- ☐ Terminate a build if it's stuck

Save **Apply**

In the **Environment** choose to provide NodeJS configurations with the version you have selected in the **Global Tool Configuration**.

Dashboard > instavote > result-build > Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment**
- Build Steps
- Post-build Actions

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

- ☐ Delete workspace before build starts
- ☐ Use secret text(s) or file(s) ?
- ☐ Provide Configuration files ?
- ☐ Add timestamps to the Console Output
- ☐ Inspect build log for published build scans
- ☒ Provide Node & npm bin/ folder to PATH

NodeJS Installation

Specify needed nodejs installation where npm installed packages will be provided to the PATH

NodeJS 22.4.0

npmrc file

- use system default -

Cache location

Default (-/npm or %APP_DATA%\npm-cache)

☐ Terminate a build if it's stuck

☐ With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Add build step

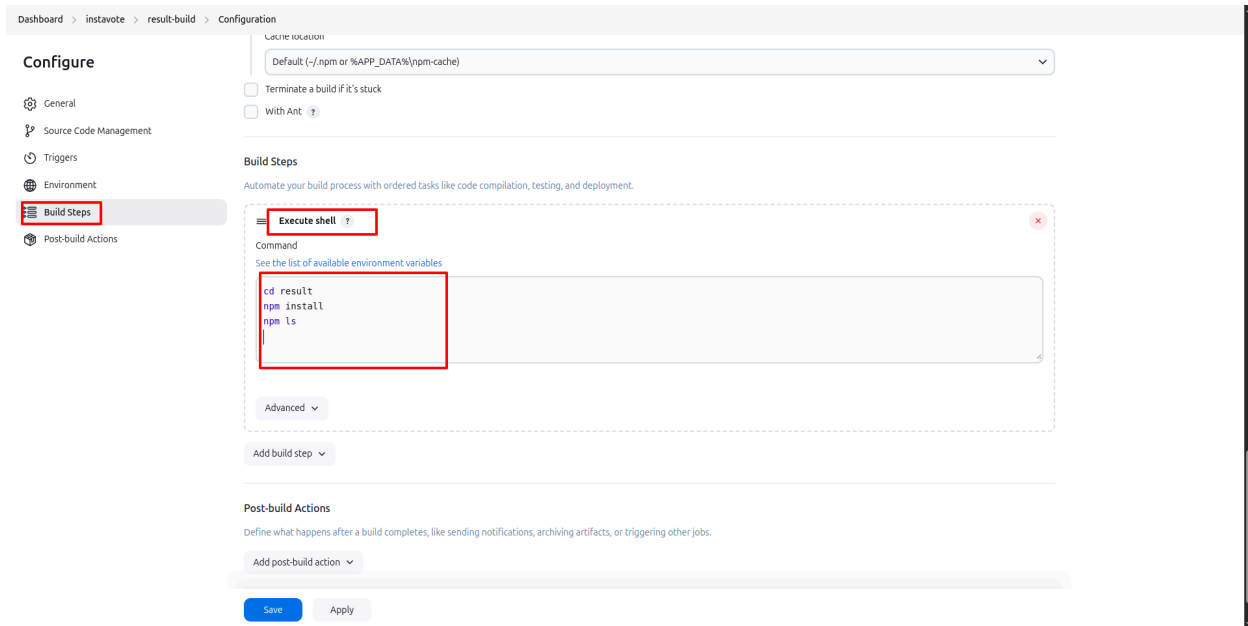
Save **Apply**

Select the **Execute shell** option from **Build steps > Add build step** and paste or type the following into the box:

```
cd result
```

```
npm install
```

```
npm ls
```



Save the project and build it.

Create another project with the name **result-test**, this time copying from **result-build**.

Change the build trigger to run after `result-build` is run.

The screenshot shows the Jenkins Configuration page for a job named 'result-test'. The 'Triggers' tab is selected in the left sidebar. The 'Triggers' section is highlighted with a red box. The 'Build after other projects are built' checkbox is checked and also highlighted with a red box. The 'Projects to watch' field contains 'result-build,'. The 'Trigger only if build is stable' radio button is selected. The 'Poll SCM' checkbox is checked. The 'Schedule' field contains 'H/2 * * * *'. The 'Save' button is highlighted with a blue box.

Dashboard > instavote > result-test > Configuration

Configure

- General
- Source Code Management
- Triggers**
- Environment
- Build Steps
- Post-build Actions

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- ☐ Trigger builds remotely (e.g., from scripts) ?
- ☒ **Build after other projects are built** ?

Projects to watch

result-build,

- ☒ Trigger only if build is stable
- ☐ Trigger even if the build is unstable
- ☐ Trigger even if the build fails
- ☐ Always trigger, even if the build is aborted

- ☐ Build periodically ?
- ☐ GitHub hook trigger for GITScm polling ?
- ☒ **Poll SCM** ?

Schedule ?

H/2 * * * * *

Would last have run at Saturday, May 3, 2025, 8:38:30 PM Coordinated Universal Time; would next run at Saturday, May 3, 2025, 8:40:30 PM Coordinated Universal Time.

- ☐ Ignore post-commit hooks ?

Save Apply

Update the command to run `npm test` instead of `npm ls`.

Build Steps

The screenshot shows the Jenkins Build Steps configuration page. The 'Execute shell' step is selected. The 'Command' field contains the following text: 'cd result', 'npm install', and 'npm test'. The 'npm test' line is highlighted with a red box. The 'Advanced...' button is visible below the command field. The 'Add build step' button is at the bottom.

Execute shell ?

Command

See [the list of available environment variables](#)

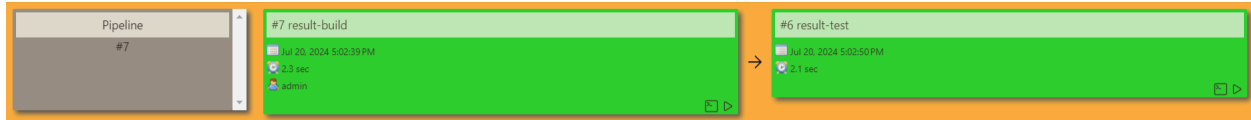
```
cd result
npm install
npm test
```

Advanced...

Add build step ▾

Create a new pipeline view, this time by choosing **result-build** as the first job to run.

After **result-build** runs, wait a few moments and refresh the page; you should see both jobs turn green in the pipeline as follows:



This completes the CI pipeline for the **result** app created with NodeJS.